Operators

Search operators

Unlock more complex operators at Expert, Master and Grandmaster Genius levels.

Arithmetic

Operator	Description
abs(x) base	Absolute value of x
add(x, y, filter = false), x + y base	Add all inputs (at least 2 inputs required). If filter = true, filter all input NaN to 0 before adding
densify(x) base	Converts a grouping field of many buckets into lesser number of only available buckets so as to make working with grouping fields computationally efficient Show less

This operator converts a grouping field with many buckets into a lesser number of only the available buckets, making working with grouping fields computationally efficient. The example below will clarify the implementation.

Example:

Say a grouping field is provided as an integer (e.g., industry: tech -> 0, airspace -> 1, ...) and for a certain date, we have instruments with grouping field values among {0, 1, 2, 99}. Instead of creating 100 buckets and keeping 96 of them empty, it is better to just create 4 buckets with values {0, 1, 2, 3}. So, if the number of unique values in x is n, densify maps those values between 0 and (n-1). The order of magnitude need not be preserved.

divide(x, y), x / y base	x / y
inverse(x) base	1 / x
log(x) base	Natural logarithm. For example: Log(high/low) uses natural logarithm of high/low ratio as stock weights.
max(x, y, ..) base	Maximum value of all inputs. At least 2 inputs are required Show less

Example:

```
max(close, vwap)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	2	1	0.01	Industry	On	Off	Verify

min(x, y ..)
base

Minimum value of all inputs. At least 2 inputs are required

Show less

Example:

```
min(close, vwap)
```

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Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	0.01	Industry	On	Off	Verify

multiply(x ,y, ... , filter=false), x * y
base

Multiply all inputs. At least 2 inputs are required. Filter sets the NaN values to 1

Show less

Examples:

```
multiply(rank(-returns), rank(volume/adv20), filter=true)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	0.01	Industry	On	Off	Verify

power(x, y)
base

x ^ y

Show less

```
1 power (returns, volume/adv20), power (returns, volume/adv20, precise=true)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	0.01	Industry	On	Off	Verify

power (x, y) operator can be used to implement popular mathematical functions. For example, sigmoid(close) can be implemented using power(x) as:

```
1 1/(1+ power(2.7182, -close)
```

Open example alpha in Simulate

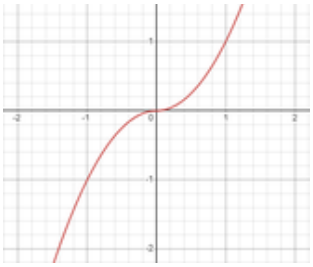
Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	1	1	1	Market	On	Off	Verify

reverse(x) base	- x
sigmoid(x) genius	Returns 1 / (1 + exp(-x))
sign(x) base	if input = NaN; return NaN
signed_power(x, y) base	x raised to the power of y such that final result preserves sign of x Show less

sign(x) * (abs(x) ^ y)

x raised to the power of y such that final result preserves sign of x. For power of 2, x ^ y will be a parabola but signed_power(x, y) will be odd and one-to-one function (unique value of x for certain value of signed_power(x, y)) unlike parabola.



Example:
If x = 3, y = 2 ⇒ abs(x) = 3 ⇒ abs(x) ^ y = 9 and sign(x) = +1 ⇒ sign(x) * (abs(x) ^ y) = signed_power(x, y) = 9
If x = -9, y = 0.5 ⇒ abs(x) = 9 ⇒ abs(x) ^ y = 3 and sign(x) = -1 ⇒ sign(x) * (abs(x) ^ y) = signed_power(x, y)

sqrt(x) base	Square root of x
subtract(x, y, filter=false), x - y base	x-y. If filter = true, filter all input NaN to 0 before subtracting
tanh(x) genius	Hyperbolic tangent of x Show less

tanh(x)

Explanation: Hyperbolic tangent of x. Output will be between -1 and 1. For very small values of x, tanh(x) = x. This is an odd and one-to-one function.

Example:

- If x = 0, tanh(x) = 0
- If x = 2, tanh(x) = 0.964
- If x = -2, tanh(x) = -0.964

Logical

Operator	Description
and(input1, input2) base	Logical AND operator, returns true if both operands are true and returns false otherwise
if_else(input1, input2, input 3) base	If input1 is true then return input2 else return input3. Show less

if_else(event_condition, Alpha_expression_1, Alpha_expression_2)

If the event condition provided is true, Alpha_expression_1 will be returned. If the event condition provided is false, Alpha_expression_2 will be returned.

We are interested in testing our hypothesis that if the stock price of a company has increased over the last 2 days, it may decrease in the future. Also, if the number of stocks bought and sold today is higher than the monthly average, then the reversion effect may be observed more profoundly.

We will implement this hypothesis by taking positions according to the difference of close price today and 3 days ago with alpha_2 using the ts_delta operator. When current volume is higher than average daily volume, we will take a larger position by multiplying by 2 to get alpha_1.

```
Event = volume > adv20;
alpha_1 = 2 * (-ts_delta(close, 3));
alpha_2 = (-ts_delta(close, 3));
if_else(event, alpha_1, alpha_2)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	0.01	Industry	On	Off	Verify

input1 < input2 base	If input1 < input2 return true, else return false
input1 <= input2 base	Returns true if input1 <= input2, return false otherwise
input1 == input2 base	Returns true if both inputs are same and returns false otherwise
input1 > input2 base	Logic comparison operators to compares two inputs
input1 >= input2 base	Returns true if input1 >= input2, return false otherwise
input1!= input2 base	Returns true if both inputs are NOT the same and returns false otherwise
is_nan(input) base	If (input == NaN) return 1 else return 0

Show less

is_nan(x) operator can be used to identify NaN values and replace them to a default value using if_else statement. For example:

```
if_else(is_nan(rank(sales)), 0.5, rank(sales))
```

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	1	1	1	Market	On	Off	Verify

In this example, in case sales value is NaN for any instrument, then the expression will replace it with the mean value of rank, that is 0.5.

not(x) base	Returns the logical negation of x. If x is true (1), it returns false (0), and if input is false (0), it returns true (1).
or(input1, input2) base	Logical OR operator returns true if either or both inputs are true and returns false otherwise

Time Series

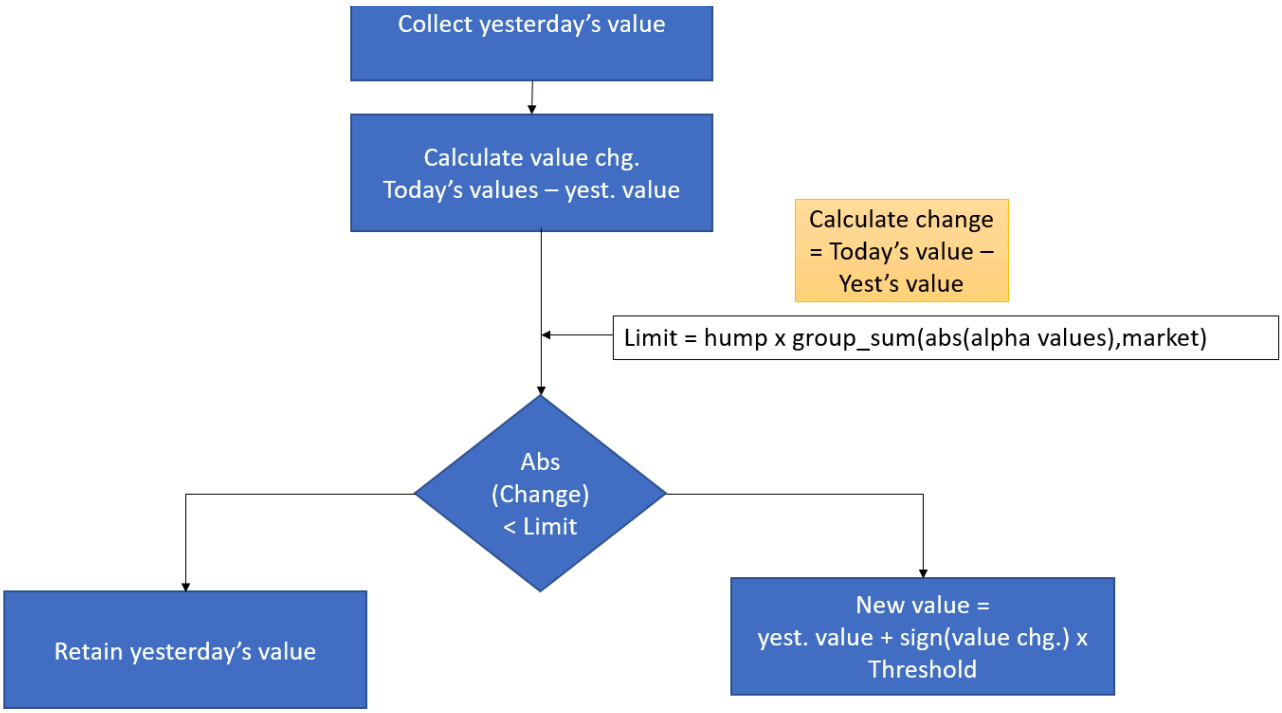
Operator	Description
days_from_last_change(x) base	Amount of days since last change of x
hump(x, hump = 0.01) base	Limits amount and magnitude of changes in input (thus reducing turnover) Show less

hump(x, hump = 0.01)

This operator limits the frequency and magnitude of changes in the Alpha (thus reducing turnover). If today's values show only a minor change (not exceeding the Threshold) from yesterday's value, the output of the hump operator stays the same as yesterday. If the change is bigger than the limit, the output is yesterday's value plus the limit in the direction of the change.

This operator may help reduce turnover and drawdown.

Flowchart of the Hump operator:




```
hump(-ts_delta(close, 5), hump = 0.00001)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	0.01	Market	On	Off	Verify

kth_element(x, d, k)
base

Returns K-th value of input by looking through lookback days. This operator can be used to backfill missing data if k=1

Show less

Returns k-th value of input by looking through lookback days while ignoring space separated scalars in ignore list. This operator is also known as **backfill** operator as it can be used to backfill missing data.

ignore parameter is used to provide list of separated scalars to ignore from counting

Example of backfill:

```
kth_element(sales/assets,252,k="1",ignore="NAN 0")
```

Open example alpha in Simulate

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	0.01	Industry	On	Off	Verify

last_diff_value(x, d)
base

Returns last x value not equal to current x value from last d days

ts_arg_max(x, d)
base

Returns the relative index of the max value in the time series for the past d days. If the current day has the max value for the past d days, it returns 0. If previous day has the max value for the past d days, it returns 1

Show less

It returns the relative index of the max value in the time series for the past d days. If the current day has the max value for the past d days, it returns 0. If previous day has the max value for the past d days, it returns 1.

Example:
If d = 6 and values for past 6 days are [6,2,8,5,9,4] with first element being today's value then max value is 9 and it is present 4 days before today. Hence, ts_arg_max(x, d) = 4

ts_arg_min(x, d)
base

Returns the relative index of the min value in the time series for the past d days; If the current day has the min value for the past d days, it returns 0; If previous day has the min value for the past d days, it returns 1.

Show less

ts_arg_min(x, d)

It returns the relative index of the min value in the time series for the past d days. If the current day has the min value for the past d days, it returns 0. If previous day has the min value for the past d days, it returns 1.

Example:
If d = 6 and values for past 6 days are [6,2,8,5,9,4] with first element being today's value then min value is 2 and it is present 1 days before today. Hence, ts_arg_min(x, d) = 1

ts_av_diff(x, d)
base

Returns x - tsmean(x, d), but deals with NaNs carefully. That is NaNs are ignored during mean computation

Show less

This operator returns x – ts_mean(x, d), but it deals with NaNs carefully

Example:
If d = 6 and values for past 6 days are [6,2,8,5,9,NaN] then ts_mean(x,d) = 6 since NaN are ignored from mean computation. Hence, ts_av_diff(x,d) = 6-6 = 0

k=1, ignore="NAN") base	value) Show less	

ts_backfill(x,lookback = d, k=1, ignore="NAN")

The ts_backfill operator replaces NaN values with the last available non-NaN value. If the input value of the data field x is NaN, the ts_backfill operator will check available input values of the same data field for the past d number of days, and output the most recent available non-NaN input value. If the k parameter is set, then the ts_backfill operator will output the kth most recent available non-NaN input value.

This operator improves weight coverage and may help to reduce drawdown risk.

Example: ts_backfill(x, 252)

- If the input value for data field x = non-NaN, then output = x
- If the input value for data field x = NaN, then output = most recent available non-NaN input value for x in the past 252 days

ts_corr(x, y, d) base	Returns correlation of x and y for the past d days Show less	

ts_corr(x, y, d)

Pearson correlation measures the linear relationship between two variables. It's most effective when the variables are normally distributed and the relationship is linear.

$$Correlation(x, y) = \frac{\sum_{i=t-d+1}^{t}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=t-d+1}^{t}(x_i - \bar{x})^2(y_i - \bar{y})^2}}$$

Example:

```
ts_corr(vwap, close, 20)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	0.01	Industry	On	Off	Verify

ts_count_nans(x ,d) base	Returns the number of NaN values in x for the past d days
ts_covariance(y, x, d) base	Returns covariance of y and x for the past d days

= false) base	sparse mode and we treat NaN as 0. In dense mode we do not. Show less

ts_decay_linear(x, d, dense = false)

Returns the linear decay on x for the past d days. Dense parameter=false means operator works in sparse mode and we treat NaN as 0. In dense mode we do not. Data smoothing techniques like linear decay reduce noise in time-series data by applying a decay factor to older observations, which helps to stabilize the dataset.

This operator improves turnover and drawdown.

Example:

- For a stock with the following prices over the last 5 days:
  - Day 0: 30 (outlier)
  - Day -1: 5
  - Day -2: 4
  - Day -3: 5
  - Day -4: 6
- The calculation would be:
  - Numerator = (30·5)+(5·4)+(4·3)+(5·2)+(6·1)=150+20+12+10+6=198
  - Denominator=5+4+3+2+1=15
  - Weighted Average=198/15=13.2
- The weighted average value of 13.2 is used instead of the outlier value of 30 for assigning weight.

ts_delay(x, d) base	Returns x value d days ago
ts_delta(x, d) base	Returns x - ts_delay(x, d)
ts_entropy(x,d) genius	For each instrument, we collect values of input in the past d days and calculate the probability distribution then the information entropy via a histogram as a result Show less

For each instrument, we collect values of the input from the past d days and calculate the probability distribution, then the information entropy via a histogram as a result. You can control the number of buckets using the buckets parameter, the default value of which is 10. To know more about entropy, refer to: Entropy information theory

Hence, output for each instrument is: log(d) - 1/d * sum_over_buckets(H[i] * log(H[i])), where H is histogram with number of buckets equal to parameter 'buckets'.

ts_mean(x, d) base	Returns average value of x for the past d days.

genius

ts_min_max_cps(x, d, f = 2) genius	Returns (ts_min(x, d) + ts_max(x, d)) - f * x. If not specified, by default f = 2
ts_min_max_diff(x, d, f = 0.5) genius	Returns x - f * (ts_min(x, d) + ts_max(x, d)). If not specified, by default f = 0.5
ts_product(x, d) base	Returns product of x for the past d days Show less

ts_product(x, d) can be used to calculate geometric mean of data fields. The geometric mean is generally a better method for averaging rates of return, growth rates of fundamentals. For example, geometric mean of daily stock returns for past 10 days can be calculated as:

```
power(ts_product(returns, 10), 1/10)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	1	1	1	Market	On	Off	Verify

ts_quantile(x,d, driver="gaussian") base	It calculates ts_rank and apply to its value an inverse cumulative density function from driver distribution. Possible values of driver (optional) are "gaussian", "uniform", "cauchy" distribution where "gaussian" is the default.
ts_rank(x, d, constant = 0) base	Rank the values of x for each instrument over the past d days, then return the rank of the current value + constant. If not specified, by default, constant = 0.
ts_regression(y, x, d, lag = 0, rettype = 0) base	Returns various parameters related to regression function Show less

ts_regression(y, x, d, lag = 0, rettype = 0)

Given a set of two variables' values (X: the independent variable, Y: the dependent variable) over a course of d days, an approximating linear function can be defined, such that sum of squared errors on this set assumes minimal value:

$$(x_i, y_i), \qquad i \in 1 \dots d, \qquad \Rightarrow \tilde{y}_i = \beta x_i + \alpha$$

$$\beta, \alpha = \arg\min \sum_i (y_i - \tilde{y}_i)^2 = \arg\min_{\alpha,\beta} \sum_i (y_i - \beta x_i - \alpha)^2.$$

Beta and Alpha in second line are OLS Linear Regression coefficients.

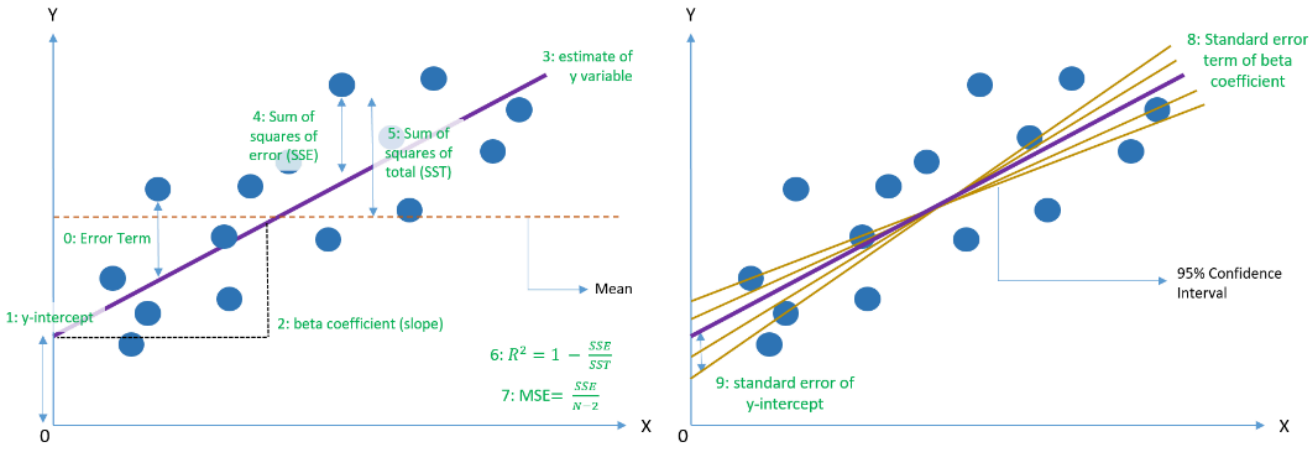
correspond to:

$$0: y_{di} - \tilde{y}_{di}, \quad 1: \alpha, \quad 2: \beta, \quad 3: \tilde{y}_{di},$$

$$4: \mathrm{SSE} = \Sigma_i (y_i - \beta x_i - \alpha)^2, \quad 5: \mathrm{SST} = \Sigma_i \left(y_i^2 - y_i \cdot n^{-1} \Sigma_j y_j\right), \quad 6: \mathrm{R}^2 = 1 - \frac{\mathrm{SSE}}{\mathrm{SST}}$$

$$7: \mathrm{s}^2 = \frac{\mathrm{SSE}}{\mathrm{N} - 2}, \quad 8: \mathrm{s}^2 \cdot \sqrt{n^{-1} + \frac{n^{-2}(n-1) \cdot \Sigma_j x_j}{\Sigma_i \left(x_i - n^{-1} \Sigma_j x_j\right)^2}}, \quad 9: \sqrt{\frac{n^{-1}(n-1) \cdot \mathrm{s}^2}{\Sigma_i \left(x_i - n^{-1} \Sigma_j x_j\right)^2}}$$

0	Error Term
1	y-intercept (α)
2	slope (β)
3	y-estimate
4	Sum of Squares of Error (SSE)
5	Sum of Squares of Total (SST)
6	R-Square
7	Mean Square Error (MSE)
8	Standard Error of β
9	Standard Error of α



Here, "di" is current day index, “n”(may differ from d) is a number of valid (x, y) tuples used for calculation. All summations are over day index, using only valid tuples.

“lag” keyword argument may be optionally specified (default value is zero) to calculate lagged regression parameters instead:

$$\tilde{y}_i = \beta x_{i-lag} + \alpha$$

- ts_regression(est_netprofit, est_netdebt, 252, lag = 0, rettype = 2)
  - Taking the data from the past 252 trading days (1 year), return the β coefficient from the equation when estimating the est_netprofit using the est_netdebt

```
ts_regression(ts_mean(volume, 2), ts_returns(close, 2), 252)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	5	Market	On	Off	Verify

ts_scale(x, d, constant = 0)
base

Returns (x - ts_min(x, d)) / (ts_max(x, d) - ts_min(x, d)) + constant. This operator is similar to scale down operator but acts in time series space

Show less

This operator returns (x – ts_min(x, d)) / (ts_max(x, d) – ts_min(x, d)) + constant
This operator is similar to scale down operator but acts in time series space

Example:
If d = 6 and values for last 6 days are [6,2,8,5,9,4] with first element being today's value,
ts_min(x,d) = 2, ts_max(x,d) = 9
ts_scale(x,d,constant = 1) = 1 + (6-2)/(9-2) = 1.57

ts_skewness(x, d)
genius

Return skewness of x for the past d days

Show less

ts_skewness(x,d)

Skewness is the third central standardized moment of the time-series vector X. Skewness can be treated as a measure of the asymmetry of the distribution of values in the time-series vector X.

$$S(X) = \frac{\mathbb{E}[(X - \mathbb{E}X)^3}{(\mathbb{E}[(X - \mathbb{E}X)^2)^{3/2}}$$

If the skewness is greater than zero, the distribution is positively skewed; if it is less than zero, it is negatively skewed; and if it is equal to zero, it is symmetric. For interpretation and analysis, focus on downside risk. Negatively skewed distributions have what statisticians call a long left tail, which for investors can mean a greater chance of extremely negative outcomes. Positive skew would mean frequent small negative outcomes, and extremely bad scenarios are not as likely.

described as being either positively skewed (meaning frequent small losses and a few extreme gains) or negatively skewed (meaning frequent small gains and a few extreme losses).

For positive skew: Mean > Median > Mode

For negative skew: Mean < Median < Mode

Example:

```
ts_skewness(returns, 60)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	0.01	Industry	On	Off	Verify

ts_std_dev(x, d) base	Returns standard deviation of x for the past d days
ts_step(1) base	Returns days' counter
ts_sum(x, d) base	Sum values of x for the past d days.
ts_target_tvr_decay(x, lambda_min=0, lambda_max=1, target_tvr=0.1) genius	Tune "ts_decay" to have a turnover equal to a certain target, with optimization weight range between lambda_min, lambda_max
ts_zscore(x, d) base	Z-score is a numerical measurement that describes a value's relationship to the mean of a group of values. Z-score is measured in terms of standard deviations from the mean: (x - tsmean(x,d)) / tsstddev(x,d). This operator may help reduce outliers and drawdown.

Cross Sectional

Operator	Description
normalize(x, useStd = false, limit = 0.0) base	Calculates the mean value of all valid alpha values for a certain date, then subtracts that mean from each element Show less

normalize(x, useStd = false, limit = 0.0)

This operator calculates the mean value of all valid alpha values for a certain date, then subtracts that mean from each element. If useStd= true, the operator calculates the standard

Example:
If for a certain date, instrument value of certain input x is [3,5,6,2], mean = 4 and standard deviation = 1.82
normalize(x, useStd = false, limit = 0.0) = [3-4,5-4,6-4,2-4] = [-1,1,2,-2]
normalize(x, useStd = true, limit = 0.0) = [-1/1.82,1/1.82,2/1.82,-2/1.82] = [-0.55,0.55,1.1,-1.1]

quantile(x, driver = gaussian, sigma = 1.0) base	Rank the raw vector, shift the ranked Alpha vector, apply distribution (gaussian, cauchy, uniform). If driver is uniform, it simply subtract each Alpha value with the mean of all Alpha values in the Alpha vector Show less

quantile(x, driver = gaussian, sigma = 1.0)

Rank the input raw Alpha vector
The ranked Alpha value would be within [0, 1]

1. Shift the ranked Alpha vector
   For every Alpha value in the ranked Alpha vector, it is shifted as: Alpha_value = 1/N + Alpha_value * (1 - 2/N), here assume there are N instruments with value in the Alpha vector. The shifted Alpha value would be within [1/N, 1-1/N]
2. Apply distribution for each Alpha value in the ranked Alpha vector using the specified driver. Driver can be one of "gaussian", "uniform", "cauchy".

Note : Sigma only affects the scale of the final value.

This operator may help reduce outliers.

Example:

```
1  quantile(close, driver = gaussian, sigma = 0.5 )
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	0.01	Market	On	Off	Verify

rank(x, rate=2) base	Ranks the input among all the instruments and returns an equally distributed number between 0.0 and 1.0. For precise sort, use the rate as 0 Show less

rank(x, rate=2):

sorting is done precisely. The default value of rate is 2.

This operator may help reduce outliers and drawdown while improving the Sharpe.

Example:

Rank(close); Rank (close, rate=0) # Sorts precisely

X = (4,3,6,10,2) => Rank(x) = (0.5, 0.25, 0.75, 1, 0)

regression_proj(y, x) genius	Conducts the cross-sectional regression on the stocks with Y as target and X as the independent variable Show less

regression_proj (Y, X)

The operator conducts the cross-sectional regression on the stocks with Y as target and X as the independent variable; Parameters 'a' and 'b' are calculated and then final output is computed for each of the stocks as (a+ (b*X))

Example:

"-returns-regression_proj (-returns, ts_stddev(returns,20))" is equivalent to regression_neut (-returns, close)

scale(x, scale=1, longscale=1, shortscale=1) base	Scales input to booksize. We can also scale the long positions and short positions to separate scales by mentioning additional parameters to the operator Show less

scale (x, scale=1, longscale=1, shortscale=1)

The operator scales the input to the book size. We can optionally tune the book size by specifying the additional parameter 'scale=booksize_value'. We can also scale the long positions and short positions to separate scales by specifying additional parameters: longscale=long_booksize and shortscale=short_booksize. The default value of each leg of the scale is 0, which means no scaling, unless specified otherwise. Scale the alpha so that the sum of abs(x) over all instruments equals 1. To scale to a different book size, use Scale(x) * booksize.

This operator may help reduce outliers.

Please check examples for the application of the same

Examples:

scale(returns, scale=4); scale (returns, scale= 1) + scale (close, scale=20); scale (returns, longscale=4, shortscale=3)

vector_neut(x, y) genius	For given vectors x and y, it finds a new vector x* (output) such that x* is orthogonal to y

vector_neut(x,y)

Input1 neutralize to input2

For given vector A (i.e., input1) and B (i.e., input2), it finds a new vector A' (i.e., output) such that A' is orthogonal to B. It calculates projection of A onto B, and then subtracts projection vector from A to find the rejection vector (i.e., A') which is perpendicular to the B.

This operator may help reduce correlation, depending on the neutralization used.

Example:

```
vector_neut(open,close)
```

Open example alpha in Simulate

Simulation Settings

Region	Universe	Language	Decay	Delay	Truncation	Neutralization	Pasteurization	NaN Handling	Unit Handling
USA	TOP3000	Fast Expression	3	1	0.01	Industry	On	Off	Verify

vector_proj(x, y) genius	Returns vector projection of x onto y.
winsorize(x, std=4) base	Winsorizes x to make sure that all values in x are between the lower and upper limits, which are specified as multiple of std.
zscore(x) base	Z-score is a numerical measurement that describes a value's relationship to the mean of a group of values. Z-score is measured in terms of standard deviations from the mean Show less

zscore(x)

Z-score is a statistical tool that indicates how many standard deviations a data point lies from the average of a group of values. Essentially, it measures how unusual a data point is in relation to the mean, making it a handy tool for understanding deviation and comparison.

The formula to calculate a Z-score is:

$$Z\text{-}score = \frac{x - mean(x)}{std(x)}$$

Where:

- x is an individual data point
- mean(x) is the average of the data set
- std(x) is the standard deviation of the data set

A Z-score tells you how many standard deviations a particular data point is from the mean. If the Z-score is positive, the data point is above the mean, and if it's negative, it's below the mean.

Z-scores may be especially useful for normalizing and comparing different data fields for different stocks or different data fields. They allow researchers to calculate the probability of a score occurring within a standard normal distribution and compare two scores that are from different samples (which may have different means and standard deviations).

This operator may help reduce outliers.

```
zscore(close)
```

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USA	TOP3000	Fast Expression	3	1	0.03	Market	On	Off	Verify

Vector

Operator	Description
vec_avg(x) base	Taking mean of the vector field x
vec_sum(x) base	Sum of vector field x

Transformational

Operator	Description
bucket(rank(x), range="0, 1, 0.1" or buckets = "2,5,6,7,10") base	Convert float values into indexes for user-specified buckets. Bucket is useful for creating group values, which can be passed to GROUP as input Show less

Bucket

Convert float values into indexes for user-specified buckets. Bucket is useful for creating group values, which can be passed to group operators as input.

If **buckets** are specified as "num_1, num_2, ..., num_N", it is converted into brackets consisting of [(num_1, num_2, idx_1), (num_2, num_3, idx_2), ..., (num_N-1, num_N, idx_N-1)]

Thus with buckets="2, 5, 6, 7, 10", the vector "-1, 3, 6, 8, 12" becomes "0, 1, 2, 4, 5"

Thus with range="0.1, 1, 0.1", the vector "0.05, 0.5, 0.9" becomes "0, 4, 8"

Note that two hidden buckets corresponding to (-inf, start] and [end, +inf) are added by default. Use the **skipBegin**, **skipEnd** parameters to remove these buckets. Use **skipBoth** to set both **skipEnd** and **skipBegin** to true.

If you want to assign all NAN values into a separate group of their own, use **NANGroup**. The index value will be one after the last bucket

Examples:

my_group = bucket(rank(volume), range="0.1,1,0.1");

group_neutralize(sales/assets, my_group)

my_group = bucket(rank(volume), buckets ="0.2,0.5,0.7", skipBoth=True, NANGroup=True);

group_neutralize(sales/assets, my_group)

trade_when(x, y, z) base	Used in order to change Alpha values only under a specified condition and to hold Alpha values in other cases. It also allows to close Alpha positions (assign NaN values) under a specified condition Show less

This operator can be used to change Alpha values only under a specified condition and to retain Alpha values in other cases. It also allows for closing Alpha positions (assigning NaN values) under a specified condition.

Trade_When (x=triggerTradeExp, y=AlphaExp, z=triggerExitExp)

If triggerExitExp > 0, Alpha = NaN.

Else if triggerTradeExp > 0, Alpha = AlphaExp;

else, Alpha = previousAlpha

This operator may help reduce correlation and reduce turnover.

Examples:

Trade_When (volume >= ts_sum(volume,5)/5, rank(-returns), -1)

If (volume >= ts_sum(volume,5)/5), Alpha = rank(-returns);

else trade previous Alpha;

exit condition is always false.

Trade_When (volume >= ts_sum(volume,5)/5, rank(-returns), abs(returns) > 0.1)

If abs(returns) > 0.1, Alpha = nan;

else trade previous Alpha.

Group

Operator	Description
group_backfill(x, group, d, std = 4.0) base	If a certain value for a certain date and instrument is NaN, from the set of same group instruments, calculate winsorized mean of all non-NaN values over last d days Show less

group_backfill(x, group, d, std = 4.0)

If a certain value for a certain date and instrument is NaN, from the set of same group instruments, calculate winsorized mean of all non-NaN values over last d days. Winsorized mean means inputs are truncated by std * stddev where stddev is the standard deviation of inputs.

Example:
If d = 4 and there are 3 instruments(i1, i2, i3) in a group whose values for past 4 days are x[i1] = [4,2,5,5], x[i2] = [7,NaN,2,9], x[i3] = [NaN,-4,2,NaN] where first element is most recent, then if we want to backfill x, we will only have to backfill x[i3]'s first element because every other instrument's first element is non-NaN.

The non-NaN values of other groups are [4,2,5,5,7,2,9,-4,2]. Mean = 3.56, Standard deviation is 3.71 and none of the item is outside the range of 3.56 – 4 * 3.71 and 3.56 + 4 * 3.71. Hence, we don't need to clip elements to those limits. Hence, Winsorized mean = backfilled value = 3.56.

For three instruments, group_backfill(x, group, d, std = 4.0) = [4,7,3.56]

group_cartesian_product(g 1, g2) genius	Merge two groups into one group. If originally there are len_1 and len_2 group indices in g1 and g2, there will be len_1 * len_2 indices in the new group.
group_extra(x, weight, group) genius	Replaces NaN values by their corresponding group means.
group_mean(x, weight, group) base	All elements in group equals to the mean Show less

group_mean(x, group) operator can be used to calculate harmonic mean of datafields. Harmonic mean gives equal weight to each value in terms of their reciprocal contribution and is considered a better method for calculating average for fundamental ratios and factors. For example, Harmonic mean of P/E ratio (price-to-earnings) for an industry can be calculated as:

```
1 1 /(group_mean(eps/close,1, industry))
```

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USA	TOP3000	Fast Expression	1	1	1	Market	On	Off	Verify

group_neutralize(x, group) base	Neutralizes Alpha against groups. These groups can be subindustry, industry, sector, country or a constant Show less

group_neutralize(x, group)

Neutralize alpha against groups. Difference between normalize and group_neutralize is in normalize, every element is subtracted by mean of all values of all instruments on that day whereas in group_neutralize, element is subtracted by mean of all values of the group of instruments that it belongs on that day.

This operator may help reduce correlation, depending on the neutralization used.

Example:
If values of field x on a certain date for 10 instruments is [3,2,6,5,8,9,1,4,8,0] and first 5 instruments belong to one group, last 5 belong to other, then mean of first group = (3+2+6+5+8)/5 = 4.8 and mean of second group = (9+1+4+8+0)/5 = 4.4. Subtracting means from instruments of respective groups gives [3-4.8, 2-4.8, 6-4.8, 5-4.8, 8-4.8, 9-4.4, 1-4.4, 4-4.4, 8-4.4, 0-4.4] = [-1.8, -2.8, 1.2, 0.2, 3.2, 4.6, -3.4, -0.4, 3.6, -4.4]

group_rank(x, group) base	Each elements in a group is assigned the corresponding rank in this group Show less

group_rank(x, group)

Group operators are a type of cross-sectional operator that compares stocks at a finer level, where the cross-sectional operation is applied within each group, rather than across the entire market. The group_rank operator allocates the stocks to their specified group, then within each group, it ranks the stocks based on their input value for data field x and returns an equally distributed number between 0.0 and 1.0.

This operator may help reduce both outliers and drawdown while reducing correlation.

Example: group_rank(x, subindustry)

- The stocks are first grouped into their respective subindustry.
- Within each subindustry, the stocks within that subindustry are ranked based on their input value for data field x and assigned an equally distributed number between 0.0 and 1.0.

base	
group_zscore(x, group) base	Calculates group Z-score - numerical measurement that describes a value's relationship to the mean of a group of values. Z-score is measured in terms of standard deviations from the mean. zscore = (data - mean) / stddev of x for each instrument within its group.

Special

Operator	Description
inst_pnl(x) genius	Generate pnl per instruments. Please note that the use of the inst_pnl() operator in an Alpha Expression is considered as utilizing the pv1 dataset (Price Volume Data for Equity) since it relies on pv1 data for calculations.